

Harry Nguyen

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SUMMARY

CS @ Georgia State (GPA 3.75, May 2027) – Google SWE intern shipped C++/TypeScript to 3B+ Chrome users; primary engineer for backend, product-facing inference APIs, and cloud infra at TiMoto AI (3-person team). Build full-stack AI products: TypeScript/React UI layers, Python/Django/FastAPI backends, and the distributed inference infrastructure underneath. Comfortable owning concept-to-launch across the stack.

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science · **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Owned backend, product-facing inference APIs, and cloud infra for a 3-person engineering team – designed, deployed, and operated **distributed production systems** end-to-end
- Built **TypeScript REST API layer** over gRPC/Python inference stack – designed the contract between product UI and the ML serving layer; maintained across 3+ concurrent clients
- Architected **vLLM inference engine with PagedAttention** for LLM serving – eliminated KV cache memory fragmentation under concurrent load; continuous batching achieving **zero OOM failures** at production traffic with **sub-50ms p99**
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Diagnosed and resolved production **gRPC deadlock** under concurrent calls via root-cause analysis – redesigned call sequencing, achieving **100% evaluation success rate** post-fix

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Architected event-driven **TypeScript/React** system with observer pattern decoupling UI state propagation – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**; reviewed by senior Chrome engineers
- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protobuf over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable channel (**3B+ users**, sub-50ms p99, 10K+ req/sec)
- Identified settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Built stateless BaaS on AWS – JWT auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**; **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck (3s+ on large datasets); redesigned with **PostgreSQL indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi

- Submitted Go CLI features and bug fixes for multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers of 24.4K+ star project
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability and correctness under concurrent operations and partial failures

SKILLS

Languages: TypeScript, Python, Go, C++, JavaScript, SQL, Java, Bash, Rust

Frontend & Full-Stack: Next.js, React, Node.js, REST APIs, observer pattern, event-driven UI

Backend & Databases: Django, FastAPI, PostgreSQL, Redis, MongoDB, gRPC, Protocol Buffers

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain,

LLM-as-a-judge

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch,

Prometheus, GCP

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor